

Multi-model databases combine relational, document, graph, and other data models, but their management remains fragmented across incompatible schemas, query languages, and adaptation mechanisms. Building on an existing category-theoretic foundation for multi-model schemas, mappings, transformations, and querying, this thesis develops methods for propagating conceptual schema changes to logical schemas, and queries, as well as a self-adaptation approach that searches for better logical configurations under changing workloads. The adaptation process combines workload-driven search with learned cost estimation, allowing candidate configurations to be compared only from query plans. The thesis contributes formal models, propagation and synchronization algorithms, and an experimental validation. Together, the results show that multi-model data management can be described coherently, automated to a substantial degree, and adapted to evolving application needs.